

Polynomials

What a zero of a polynomial means, how graphs show those zeroes, how the zeroes are linked to the coefficients, and how one polynomial divides another, with worked examples and practice questions.

Introduction

In Class 9 you met polynomials in one variable, expressions like $2x + 3$, $x^2 - 3x + 2$ and $3x^3 - x + 1$, where the powers of x are whole numbers. In this chapter we look at those polynomials more closely. The main question is: for which values of x does the polynomial become zero? Those values are called the **zeroes** of the polynomial, and they tell us a great deal about its graph and its coefficients.

We will also see how to divide one polynomial by another. That division helps us find zeroes and check our factorisation.

Zero of a Polynomial

A value of x that makes the whole polynomial equal to zero is called a zero of that polynomial. It is not the same as the value of the polynomial, it is the input that gives output zero.

Zero of a polynomial

A real number k is a **zero** of a polynomial $p(x)$ if

$$p(k) = 0$$

For example, for $p(x) = x - 2$, we have $p(2) = 2 - 2 = 0$, so 2 is a zero of $p(x)$.

For $p(x) = x^2 - 3x + 2$, $p(1) = 1 - 3 + 2 = 0$ and $p(2) = 4 - 6 + 2 = 0$, so both 1 and 2 are zeroes.

A few points that come up often in exams:

- The **degree** of a polynomial is the highest power of x with a non-zero coefficient. A linear polynomial has degree 1 and a quadratic polynomial has degree 2.
- A polynomial of degree n can have at most n zeroes. Thus a linear polynomial has at most one zero and a quadratic polynomial has at most two.
- A non-zero constant polynomial (like $p(x) = 5$) has no zero at all, because $p(x)$ is never zero.

Do not confuse a zero of a polynomial with the point where the graph meets the y -axis. The y -intercept is $p(0)$; a zero is where $p(x) = 0$, which is where the graph meets the x -axis.

Geometric Meaning of Zeroes

The zeroes of a polynomial have a simple picture on a graph. If you draw $y = p(x)$, each zero is an x -coordinate where the graph cuts or touches the x -axis (where $y = 0$). This is why the number of zeroes matches the number of times the graph meets the x -axis.

Geometric meaning

The zeroes of a polynomial $p(x)$ are the x -coordinates of the points where the graph $y = p(x)$ intersects the x -axis.

What this looks like for each type:

- **Linear polynomial** (degree 1, e.g. $y = x + 2$): the graph is a straight line. It cuts the x -axis at exactly one point, so there is exactly one zero.
- **Quadratic polynomial** (degree 2, e.g. $y = ax^2 + bx + c$, $a \neq 0$): the graph is a parabola opening upwards (if $a > 0$) or downwards (if $a < 0$). It can meet the x -axis in three ways:
 - in two places, two distinct zeroes,
 - at one place where it just touches the axis, one repeated zero (two equal zeroes),
 - nowhere at all, no real zero, because the whole parabola lies above or below the axis.

This picture is useful because it tells you at a glance how many real zeroes to expect before you do any algebra. If a question gives you a graph, count the x-intercepts, that is the number of zeroes.

Relationship between Zeroes and Coefficients

Once you know the zeroes of a polynomial, you can connect them directly to its coefficients. This saves time in exams, you can find the sum and product of zeroes without actually solving the equation, or you can build a polynomial when the zeroes are given.

Linear polynomial

Linear polynomial

For $p(x) = ax + b$, $a \neq 0$, if k is the zero, then

$$k = -b / a$$

This comes straight from solving $ax + b = 0$.

Quadratic polynomial

Quadratic polynomial

For $p(x) = ax^2 + bx + c$, $a \neq 0$, let α and β be its zeroes. Then

$$\alpha + \beta = -b / a \quad \text{and} \quad \alpha\beta = c / a$$

That is, the sum of the zeroes is minus the coefficient of x divided by the coefficient of x^2 , and the product of the zeroes is the constant term divided by the coefficient of x^2 . The minus sign in the sum is the part students most often forget.

These two relations also let you form a quadratic polynomial when α and β are known:

$$p(x) = k \cdot [x^2 - (\alpha + \beta)x + \alpha\beta]$$

where k is any non-zero real number. Usually we take $k = 1$ to get the simplest polynomial.

Worked Example 1, Find zeroes and verify the relationship

Find the zeroes of $p(x) = x^2 - 2x - 8$ and verify the relationship between zeroes and coefficients.

Factorise by splitting the middle term. We need two numbers whose product is -8 and sum is -2 . They are -4 and 2 .

$$x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = x(x - 4) + 2(x - 4) = (x - 4)(x + 2)$$

So $p(x) = 0$ gives $x - 4 = 0$ or $x + 2 = 0$, so **$\alpha = 4$, $\beta = -2$** .

Now check against the coefficients. Here $a = 1$, $b = -2$, $c = -8$.

$$\alpha + \beta = 4 + (-2) = 2 \quad \text{and} \quad -b/a = -(-2)/1 = 2 \quad \checkmark$$

$$\alpha\beta = 4 \times (-2) = -8 \quad \text{and} \quad c/a = -8/1 = -8 \quad \checkmark$$

Worked Example 2, Form a polynomial from given zeroes

Find a quadratic polynomial whose zeroes are 3 and $-5/2$.

Let $\alpha = 3$ and $\beta = -5/2$.

$$\alpha + \beta = 3 - 5/2 = 1/2 \quad \alpha\beta = 3 \times (-5/2) = -15/2$$

A quadratic polynomial with these zeroes is

$$p(x) = x^2 - (\alpha + \beta)x + \alpha\beta = x^2 - (1/2)x - 15/2$$

To clear the fractions, multiply by 2 (any non-zero k gives a valid polynomial):

$$p(x) = 2x^2 - x - 15$$

This is the simplest polynomial with integer coefficients having 3 and $-5/2$ as zeroes. Any multiple of it, like $4x^2 - 2x - 30$, would also be correct.

Summary

- A real number k is a **zero** of $p(x)$ if $p(k) = 0$. A polynomial of degree n has at most n zeroes.
- Geometrically, the real zeroes are the x -coordinates where $y = p(x)$ meets the x -axis. A quadratic polynomial can have 0 , 1 or 2 real zeroes.
- For $p(x) = ax + b$, zero $= -b/a$.
- For $p(x) = ax^2 + bx + c$ with zeroes α , β : $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$.

- A quadratic with zeroes α, β is $k[x^2 - (\alpha + \beta)x + \alpha\beta]$ for any $k \neq 0$.
- For $p(x) = ax^3 + bx^2 + cx + d$ with zeroes α, β, γ : $\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$, $\alpha\beta\gamma = -d/a$.

Practice Questions

1. Find the zeroes of the quadratic polynomial $x^2 - 5x + 6$ and verify the relationship between zeroes and coefficients.

2. Find a quadratic polynomial whose zeroes are -3 and 4 . Verify that the sum and product of zeroes match $-b/a$ and c/a .

3. If α and β are the zeroes of $p(x) = 2x^2 + 5x - 3$, find the value of $\alpha + \beta + \alpha\beta$ without finding the zeroes individually.

4. The graph of $y = p(x)$ is a parabola that does not meet the x -axis at all. How many zeroes does $p(x)$ have? What does this tell you about the discriminant?

5. 3 is a zero of the polynomial $2x^3 - 5x^2 - 4x + 3$. Find the other zeroes.